

Logging using SLF4J

Exercise 1: Logging Error Messages and Warning Levels

```
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

public class LoggingExample {
    private static final Logger logger = LoggerFactory.getLogger(LoggingExample.class);

    public static void main(String[] args) {
        logger.error("This is an error message");
        logger.warn("This is a warning message");
    }
}
```

Output:

```
2026-07-06 06:57:44 [LoggingExample.main()] ERROR LoggingExample - This is an error
message
2026-07-06 06:57:44 [LoggingExample.main()] WARN  LoggingExample - This is a warning
message
```

Exercise 2: Parameterized Logging

```
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

public class ParameterizedLoggingExample {
    private static final Logger logger = LoggerFactory.getLogger(ParameterizedLoggingExample.class);

    public static void main(String[] args) {
        String username = "Deepak";
        int loginAttempts = 3;

        logger.info("User {} logged in successfully", username);
        logger.warn("User {} has {} failed login attempts", username, loginAttempts);
        logger.error("Login blocked for user {} after {} attempts", username, loginAttempts);
    }
}
```

```

slf4j.Logger;
slf4j.LoggerFactory;

ParameterizedLoggingExample {
    static final Logger logger = LoggerFactory.getLogger(ParameterizedLoggingExample.class);

    public void main(String[] args) {
        username = "Deepak";
        loginAttempts = 3;

        info("User {} logged in successfully", username);
        warn("User {} has {} failed login attempts", username, loginAttempts);
        error("Login blocked for user {} after {} attempts", username, loginAttempts);
    }
}

```

Output:

```

2026-07-06 07:01:25 [ParameterizedLoggingExample.main()] INFO
ParameterizedLoggingExample - User Deepak logged in successfully
2026-07-06 07:01:25 [ParameterizedLoggingExample.main()] WARN
ParameterizedLoggingExample - User Deepak has 3 failed login attempts
2026-07-06 07:01:25 [ParameterizedLoggingExample.main()] ERROR
ParameterizedLoggingExample - Login blocked for user Deepak after 3 attempts

```

Exercise 3: Using Different Appenders

```

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

public class AppenderExample {
    private static final Logger logger = LoggerFactory.getLogger(AppenderExample.class);

    public static void main(String[] args) {
        logger.debug("Debug: initializing application");
        logger.info("Info: application started successfully");
        logger.warn("Warning: low disk space detected");
        logger.error("Error: failed to connect to database");
    }
}

```

Output:

```

21:20:10.045 [main] DEBUG AppenderExample - Debug: initializing application
21:20:10.052 [main] INFO AppenderExample - Info: application started successfully
21:20:10.058 [main] WARN AppenderExample - Warning: low disk space detected
21:20:10.064 [main] ERROR AppenderExample - Error: failed to connect to database

```